

Charles Emmanuel A. Cruz

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Professional Experience

Department of Science and Technology – Planning and Evaluation Service

Taguig City

Backend Developer Intern

January – April 2026

- Developed and maintained RESTful APIs to support backend services within a microservices architecture, enabling modular and scalable system components.
- Implemented Apache Kafka for asynchronous message queueing and event-driven communication between microservices, enabling service decoupling.
- Collaborated with fellow interns and senior developers using Agile practices, participating in daily stand-ups, sprint planning, code reviews, and iterative feature delivery.
- Built and integrated backend services that streamlined data exchange across multiple microservices, improving service interoperability.
- Assisted in testing, debugging, and optimizing backend endpoints to ensure reliability and performance.
- Contributed to technical documentation, improving maintainability and conducting knowledge transfers for fellow interns.

Education

De La Salle University – Dasmariñas

Dasmariñas, Cavite

BS Information Technology

August 2026

Awards: Summa Cum Laude, DOST-SEI Merit Scholar, Champion – IT Quiz Bee, Delegate – DOST Scholars' Research Camp

Technical Skills

Programming Languages: Java, JavaScript, TypeScript, C#, Python

Backend Development: Node.js, REST APIs, Microservices Architecture

Frameworks: Express.js, ASP.NET

Messaging Systems: Apache Kafka

Frontend: HTML5, CSS3, React

Databases: PostgreSQL, MongoDB, Microsoft SQL Server

DevOps & Tools: Docker, Podman, Kubernetes, Git, Postman

Platforms & Technologies: WordPress, cPanel, Figma, Google Apps Script

Projects & Relevant Coursework

An IoT-Based Bus Tracking and Passenger Monitoring System for Jasper Jean Services Inc.

April 2026

- Developed an IoT-based bus tracking system integrating GPS with backend services for real-time data transmission.
- Integrated hardware components with a custom web dashboard using MQTT and REST APIs, enabling real-time data transmission and system synchronization.
- Implemented a YOLO-based computer vision model (Ultralytics) for real-time passenger counting, processing data before transmission to the dashboard.

Automated CI/CD Pipeline Implementation | Cisco DevNet Laboratory

December 2025

- Deployed a containerized Jenkins CI/CD environment via Docker to automate integration with a GitHub repository, triggering builds upon code modifications to reduce integration errors.
- Configured automated testing jobs within Jenkins to continuously analyze code and verify deployment readiness.

Heraldo Filipino Website Redesign (www.heraldofilipino.org)

October 2024

- Designed a modern and minimalist user interface for the official student publication's website.
- Utilized a combination of WordPress themes and plugins to improve overall user experience for site visitors.

Certifications & Certificates

- IT Specialist – Databases (Certiport)
- DevNet Associate (Cisco) | AI Fundamentals (IBM SkillsBuild) | PCAP: Programming Essentials in Python (OpenEDG)